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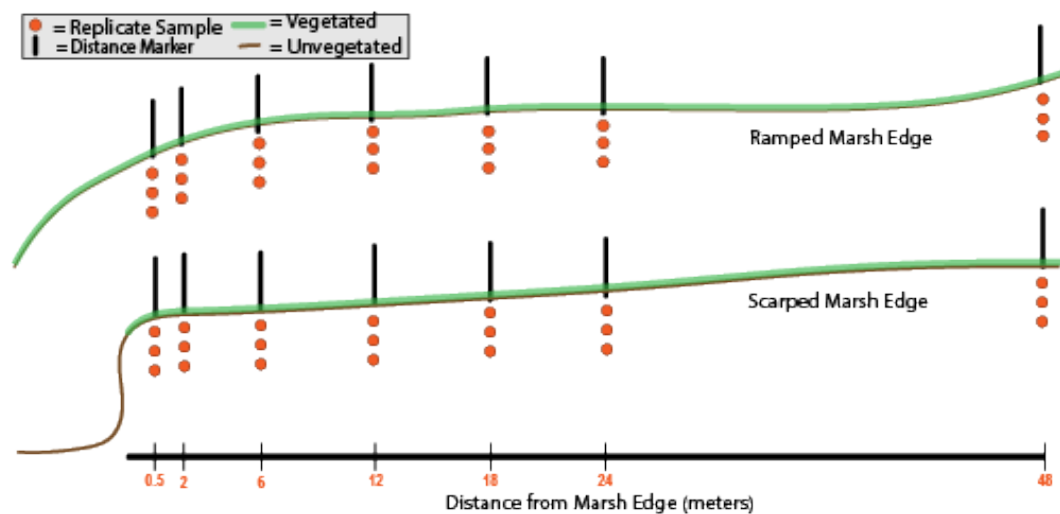


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Collection Interval Length Over Time Across Sites

Dashed line = nominal 28-day target

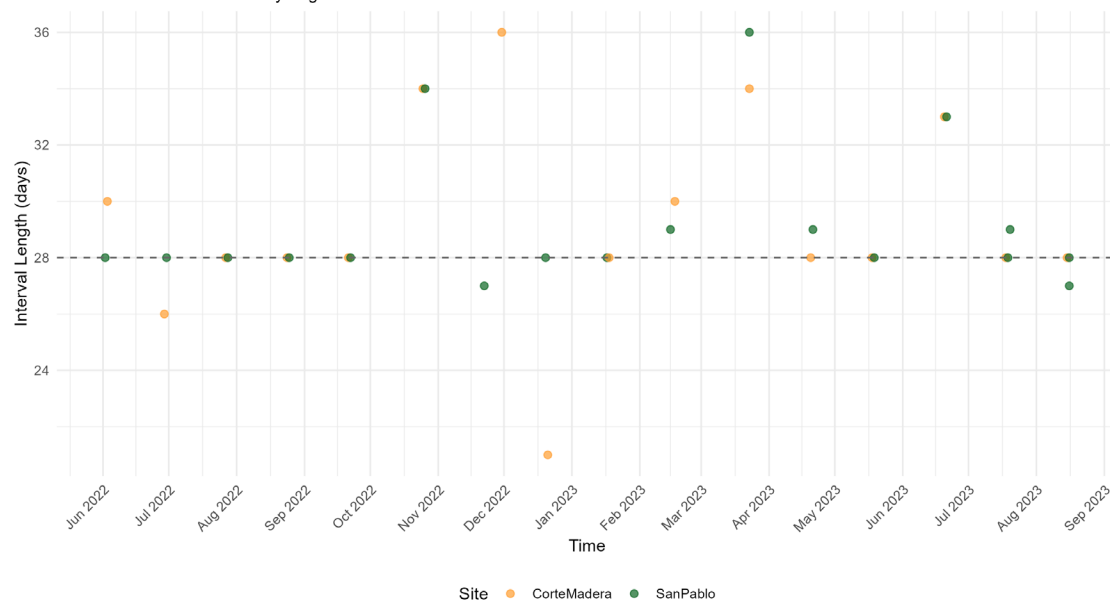
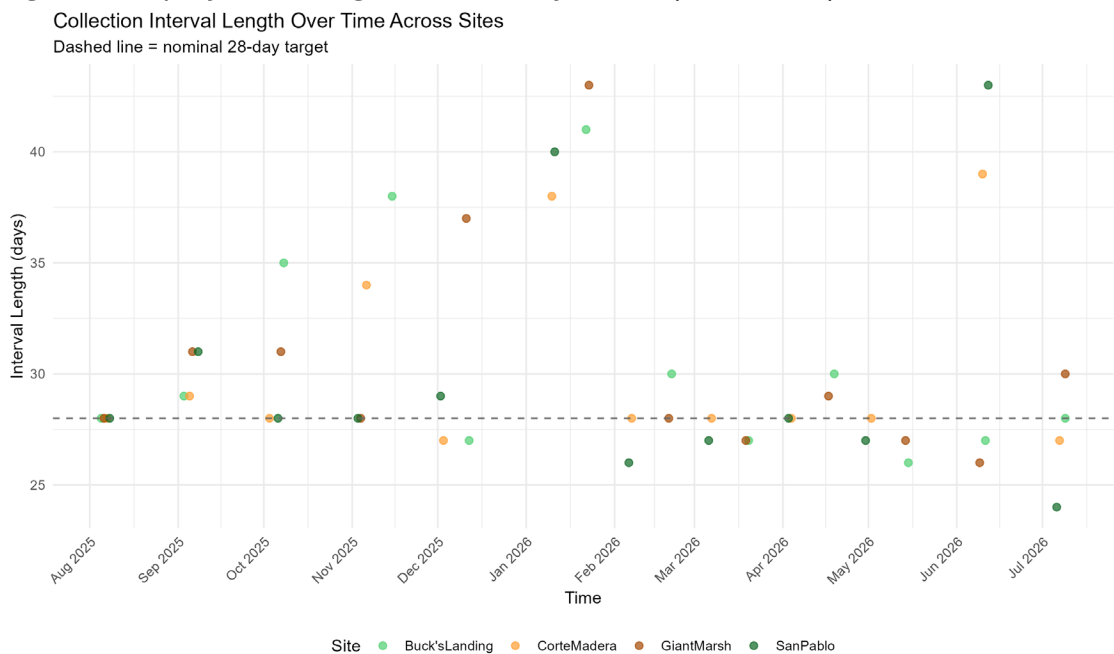


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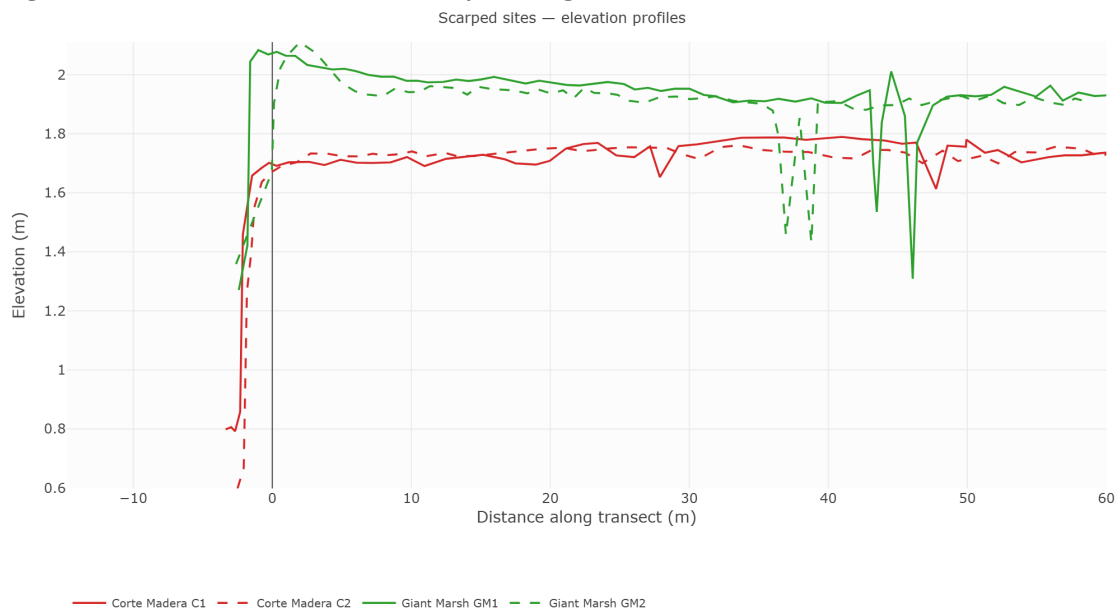
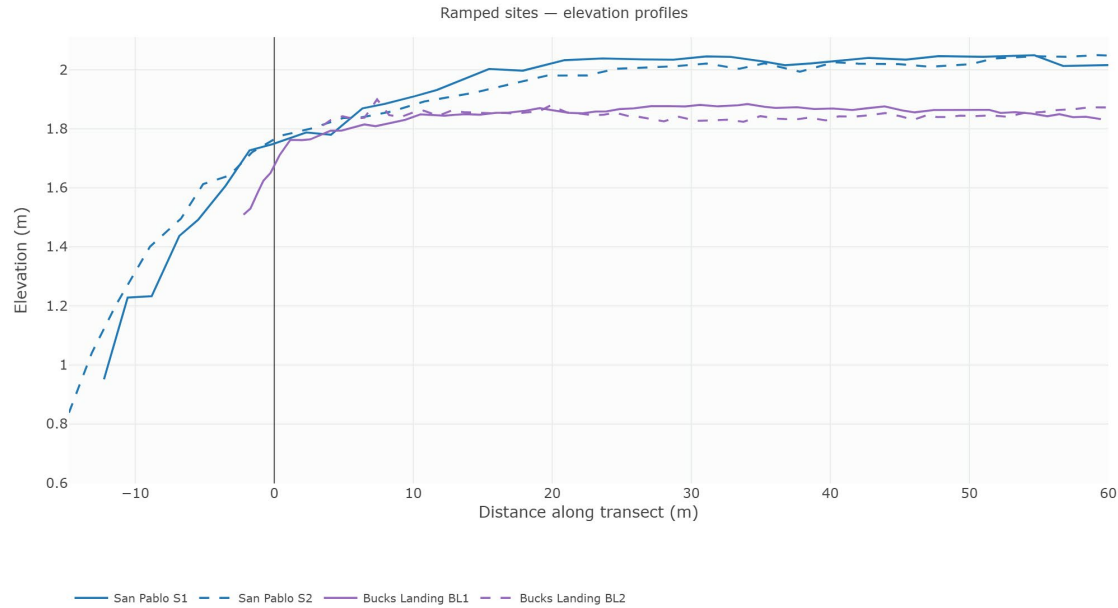


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4.1 Sediment Deposition Across Ramped and Scarped Marsh Edges

Marsh Edge Typology Level

Table 1. GLMM of Sediment Deposition Across Marsh Edge Types

Term	Estimate	SE	z	p	exp(β)	95% CI low	95% CI high
(Intercept)	1.659	0.219	7.567	< .001	5.251	3.417	8.070
marsh_edge_typeRamped	-0.466	0.309	-1.507	.132	0.627	0.342	1.150

Type II Wald χ^2 -- marsh_edge_type: $\chi^2(1) = 2.27$, $p = .132$

Random intercept (Dist): variance = 0.000, SD = 0.000 [6 groups]

Random intercept (Loc_ID): variance = 1.052, SD = 1.026 [48 groups]

N = 499 observations; AIC = 2523.5

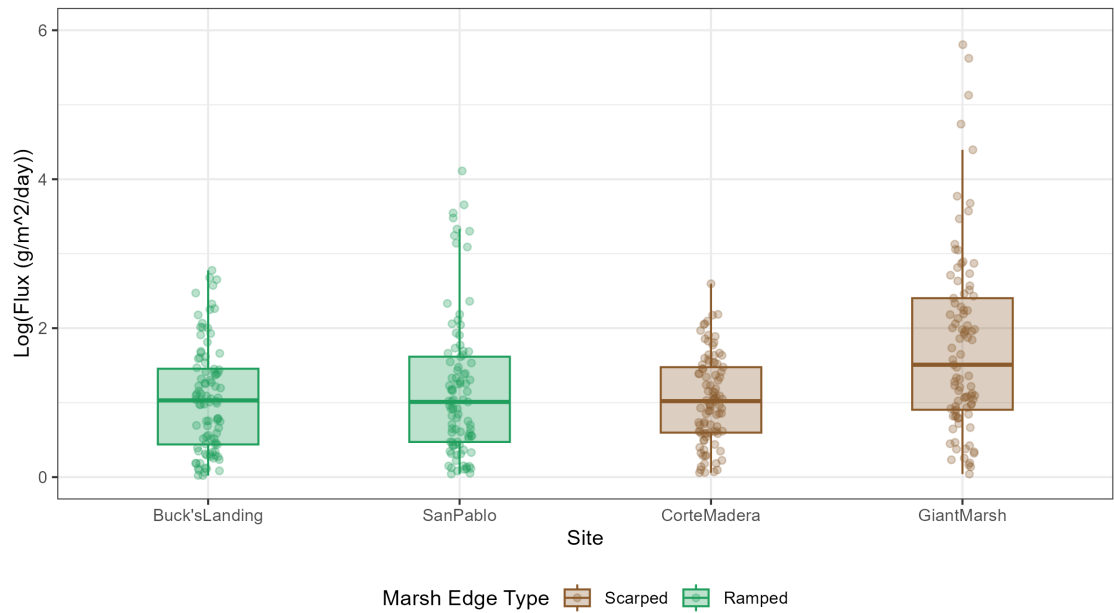
Marginal $R^2 = 0.076$; conditional $R^2 = NA$ (NA = a random-effect variance was estimated at ~ 0)

Formula: Flux ~ marsh_edge_type + (1 | Dist) + (1 | Loc_ID)

Family: Gamma (log link) -- exp(β) = multiplicative change in Flux; 95% CI = Wald

Site Level

Figure 8. Sediment Deposition (Flux) Across Site Level of Ramped and Scarped Sites (2025-
Sediment Deposition Rates Across Sites Between Scarped and Ramped Marsh Edge
Sites (Averaged)



2026)

Table 2. Statistical Table of Sediment Deposition (Flux) Across Site (2025-2026)

Site	n	Mean	SD	SE	Median	Min	Max	Skewness	Kurtosis	Shapiro_p	Shapiro_p_display
GiantMarsh	114	58.959	199.090	18.646	3.671	0.111	1,095.227	4.156	19.546	0.000	< .001
CorteMadera	125	2.695	2.282	0.204	2.082	0.006	13.421	1.518	6.308	0.000	< .001
SanPablo	126	4.573	8.609	0.767	1.811	0.077	61.001	3.914	20.404	0.000	< .001
Buck'sLanding	134	2.893	3.063	0.265	1.675	0.110	16.058	2.205	8.092	0.000	< .001

Tran-Site Level

Figure 9. Sediment Deposition (Flux) Across Transect Nested in Site Level of Ramped and Scarped Sites (2025-2026)

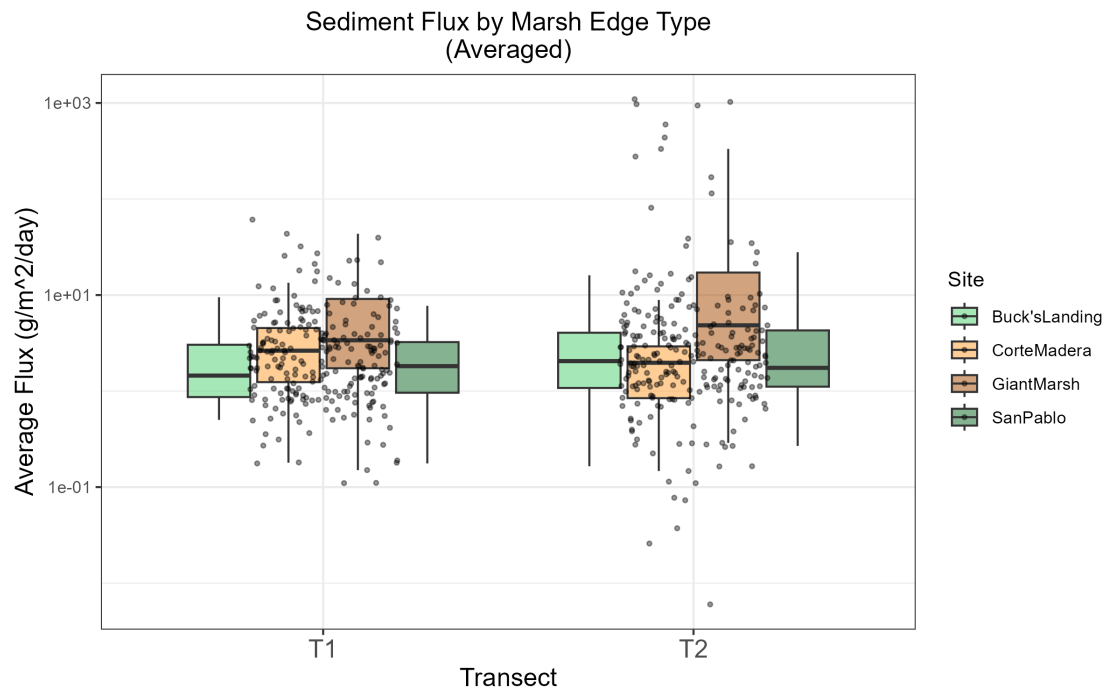


Table 3. GLMM of Sediment Deposition Across Transect Nested in Site Level of Ramped and Scarped Sites (2025-2026)

Term	Estimate	SE	z	p	exp(β)	95% CI low	95% CI high
(Intercept)	0.813	0.349	2.331	.020	2.254	1.138	4.466
TranT2	0.405	0.493	0.822	.411	1.500	0.571	3.942
SiteCorteMadera	0.247	0.495	0.499	.618	1.280	0.485	3.378
SiteGiantMarsh	1.088	0.495	2.197	.028	2.969	1.125	7.839
SiteSanPablo	0.503	0.494	1.017	.309	1.653	0.628	4.356
TranT2:SiteCorteMadera	-0.757	0.699	-1.083	.279	0.469	0.119	1.846
TranT2:SiteGiantMarsh	0.683	0.702	0.972	.331	1.979	0.499	7.841
TranT2:SiteSanPablo	-0.296	0.699	-0.423	.672	0.744	0.189	2.928

Type II Wald χ^2 -- Tran: $\chi^2(1) = 1.56$, $p = .212$

Type II Wald χ^2 -- Site: $\chi^2(3) = 24.07$, $p < .001$

Type II Wald χ^2 -- Tran:Site: $\chi^2(3) = 4.40$, $p = .221$

Random intercept (Dist): variance = 0.000, SD = 0.000 [6 groups]

Random intercept (Loc_ID): variance = 0.642, SD = 0.801 [48 groups]

N = 499 observations; AIC = 2514.5

Marginal $R^2 = 0.398$; conditional $R^2 = NA$ (NA = a random-effect variance was estimated at ~ 0)

Formula: Flux ~ Tran + Site + Tran * Site + (1 | Dist) + (1 | Loc_ID)

Family: Gamma (log link) -- exp(β) = multiplicative change in Flux; 95% CI = Wald

Z star level per site

Figure 10. Sediment Deposition (Flux) Across z^* (Relative Elevation) Within Ramped and Scarped Sites (2025-2026)

Daily Sediment Flux vs. Elevation (z^*)
Across 2 Ramped and 2 Scarped Marshes,
2025-2026 (Averaged)

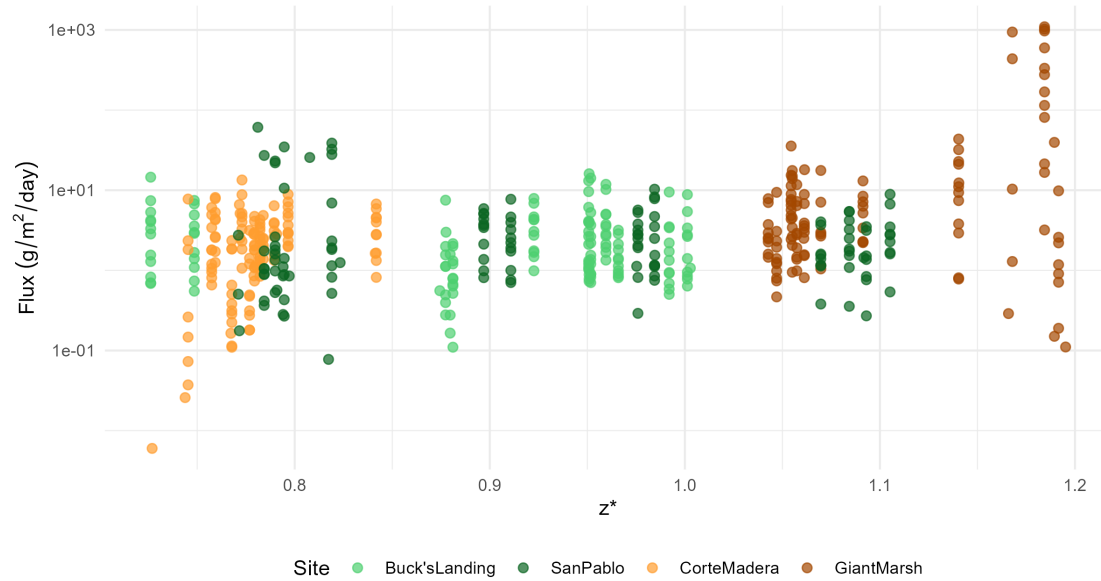


Table 4. GLMM of Sediment Deposition Across z^* (Relative Elevation) Within Ramped and Scarped Sites (2025-2026)

Term	Estimate	SE	z	p	exp(β)	95% CI low	95% CI high
(Intercept)	-3.458	1.035	-3.340	< .001	0.031	0.004	0.240
z_star	5.331	1.067	4.996	< .001	206.620	25.518	1,673.013
marsh_edge_typeRamped	4.924	1.843	2.671	.008	137.597	3.710	5,102.771
z_star :marsh_edge_typeRamped	-5.876	1.966	-2.990	.003	0.003	0.000	0.132

Type II Wald χ^2 -- z_star : $\chi^2(1) = 16.12$, $p < .001$

Type II Wald χ^2 -- marsh_edge_type: $\chi^2(1) = 4.71$, $p = .030$

Type II Wald χ^2 -- z_star :marsh_edge_type: $\chi^2(1) = 8.94$, $p = .003$

Random intercept (Site): variance = 0.000, SD = 0.000 [4 groups]

Random intercept (Loc_ID): variance = 0.671, SD = 0.819 [48 groups]

Random intercept (Study.Phase): variance = 0.485, SD = 0.696 [12 groups]

N = 499 observations; AIC = 2293.5

Marginal $R^2 = 0.485$; conditional $R^2 = NA$ (NA = a random-effect variance was estimated at ~0)

Formula: Flux ~ z_star * marsh_edge_type + (1 | Site) + (1 | Loc_ID) + (1 | Study.Phase)

Family: Gamma (log link) -- exp(β) = multiplicative change in Flux; 95% CI = Wald

Distance Level Per Site

Figure 11. Sediment Deposition (Flux) Across Distances from Marsh Edge Within Ramped and Scarped Sites (2025-2026)

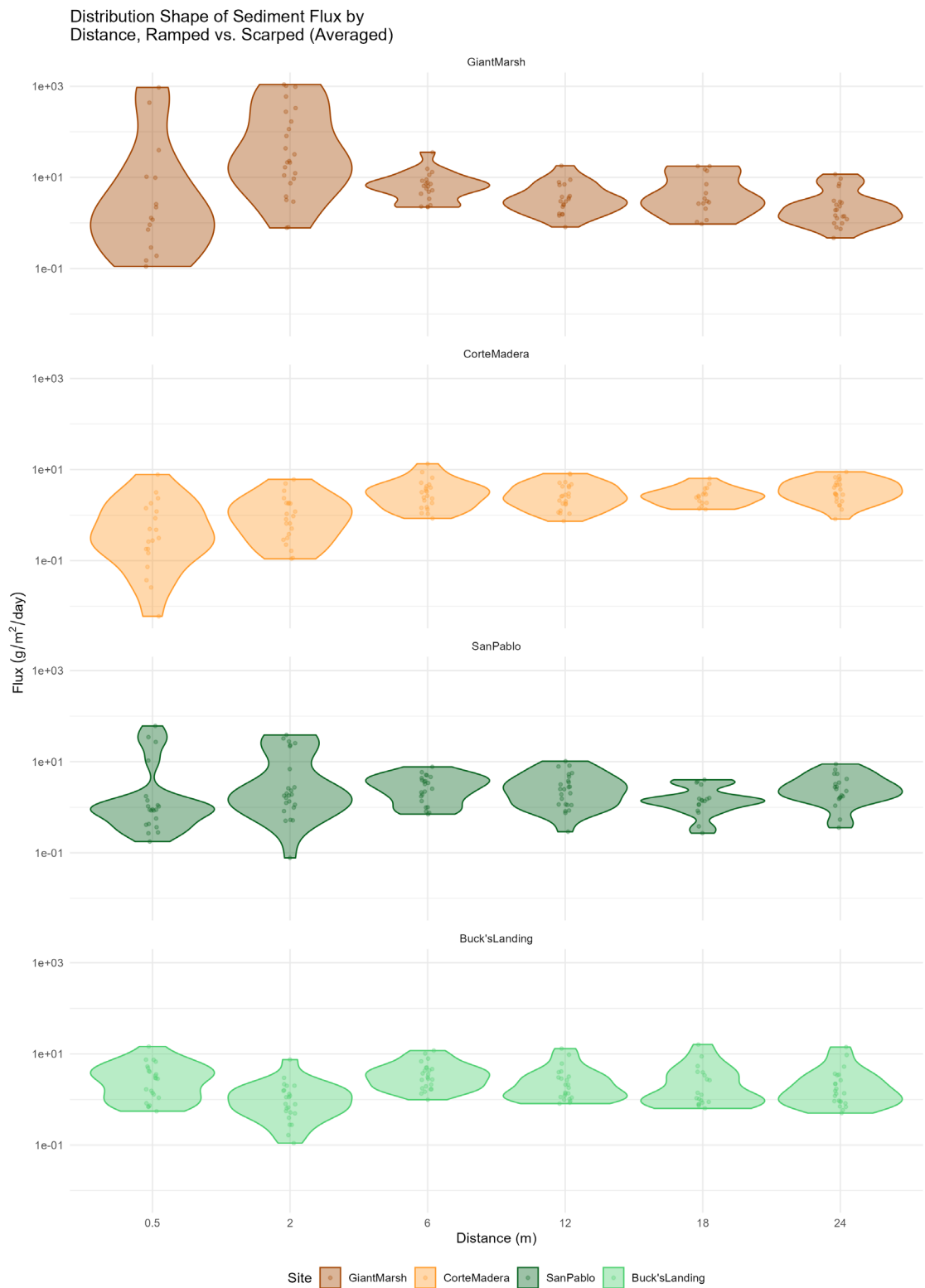


Table 5. Statistical Table of Sediment Deposition (Flux) Across Distances within Giant Marsh Site (2025-2026)

Distance (m)	n	Mean	SD	SE	Median	Min	Max	Skewness	Kurtosis	Shapiro_p
0.500	15	96.480	258.959	66.863	1.290	0.111	941.316	2.705	8.914	< .001
2.000	24	202.995	349.115	71.263	22.055	0.779	1,095.227	1.757	4.536	< .001
6.000	20	8.018	7.427	1.661	6.417	2.217	35.596	2.719	10.694	< .001
12.000	19	4.385	4.045	0.928	2.987	0.811	18.041	2.198	7.860	< .001
18.000	15	6.375	6.289	1.624	2.992	0.948	17.644	0.951	2.180	0.001
24.000	21	2.998	3.070	0.670	1.909	0.467	11.742	1.719	4.822	< .001

Flux units: g/m²/day

Kurtosis is RAW (not excess) -- a normal distribution reads ~3, not 0.

Shapiro_p < .05 indicates a significant departure from normality; NA where n < 3 or n > 5000.

Kruskal-Wallis across Distance (m) within GiantMarsh: statistic = 33.59, p < .001

Table 6. Statistical Table of Sediment Deposition (Flux) Across Distances within Corte Madera Site (2025-2026)

Distance (m)	n	Mean	SD	SE	Median	Min	Max	Skewness	Kurtosis	Shapiro_p
0.500	19	1.106	1.836	0.421	0.313	0.006	7.761	2.736	10.264	< .001
2.000	22	1.433	1.581	0.337	0.881	0.110	6.082	1.707	5.214	< .001
6.000	22	3.602	2.934	0.626	2.909	0.844	13.421	1.976	6.959	< .001
12.000	23	3.134	2.052	0.428	2.569	0.736	8.123	1.071	3.404	0.010
18.000	16	2.840	1.363	0.341	2.577	1.335	6.402	1.209	3.994	0.051
24.000	23	3.809	2.100	0.438	2.976	0.818	8.890	0.698	2.723	0.220

Flux units: g/m²/day

Kurtosis is RAW (not excess) -- a normal distribution reads ~3, not 0.

Shapiro_p < .05 indicates a significant departure from normality; NA where n < 3 or n > 5000.

Kruskal-Wallis across Distance (m) within CorteMadera: statistic = 41.27, p < .001

Table 7. Statistical Table of Sediment Deposition (Flux) Across Distances within San Pablo Site (2025-2026)

Distance (m)	n	Mean	SD	SE	Median	Min	Max	Skewness	Kurtosis	Shapiro_p
0.500	21	6.986	15.432	3.367	0.887	0.176	61.001	2.558	8.603	< .001
2.000	24	8.364	12.190	2.488	1.865	0.077	38.703	1.335	3.171	< .001
6.000	22	3.045	1.906	0.406	2.890	0.705	7.719	0.616	2.726	0.178
12.000	24	3.160	2.638	0.538	2.494	0.291	10.297	1.266	3.789	0.003
18.000	16	1.737	1.186	0.296	1.417	0.271	4.027	0.831	2.366	0.016
24.000	19	3.059	2.216	0.508	2.506	0.356	8.902	1.151	3.739	0.038

Flux units: g/m²/day

Kurtosis is RAW (not excess) -- a normal distribution reads ~3, not 0.

Shapiro_p < .05 indicates a significant departure from normality; NA where n < 3 or n > 5000.

Kruskal-Wallis across Distance (m) within SanPablo: statistic = 12.27, p = .031

Table 8. Statistical Table of Sediment Deposition (Flux) Across Distances within Buck's Landing Site (2025-2026)

Distance (m)	n	Mean	SD	SE	Median	Min	Max	Skewness	Kurtosis	Shapiro_p
0.500	23	3.512	3.250	0.678	2.897	0.551	14.550	1.829	6.757	< .001
2.000	23	1.315	1.539	0.321	0.799	0.110	7.510	3.002	12.614	< .001
6.000	23	3.986	2.847	0.594	3.057	0.987	11.849	1.406	4.289	0.002
12.000	23	2.907	3.023	0.630	1.806	0.810	13.120	2.247	7.471	< .001
18.000	19	3.024	3.798	0.871	1.266	0.637	16.058	2.427	8.526	< .001
24.000	23	2.638	3.224	0.672	1.366	0.502	14.176	2.514	8.840	< .001

Flux units: g/m²/day

Kurtosis is RAW (not excess) -- a normal distribution reads ~3, not 0.

Shapiro_p < .05 indicates a significant departure from normality; NA where n < 3 or n > 5000.

Kruskal-Wallis across Distance (m) within Buck'sLanding: statistic = 25.29, p < .001

Temporally

Figure 12. Sediment Deposition (Flux) Across July 2025 -October 2025 Within Ramped and Scarped Sites

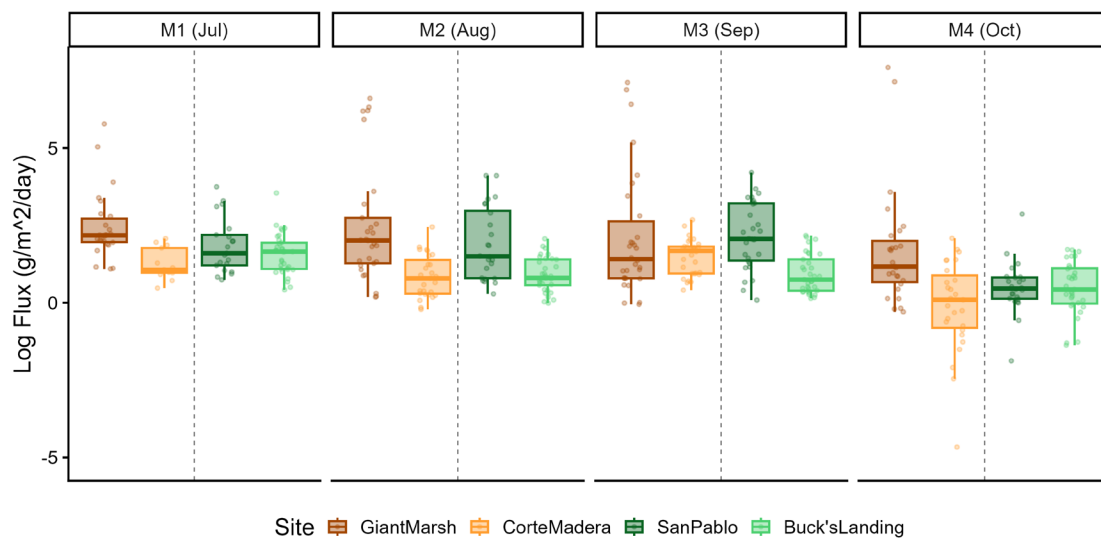


Figure 13. Sediment Deposition (Flux) Across November 2025 -February 2026 Within Ramped and Scarped Sites

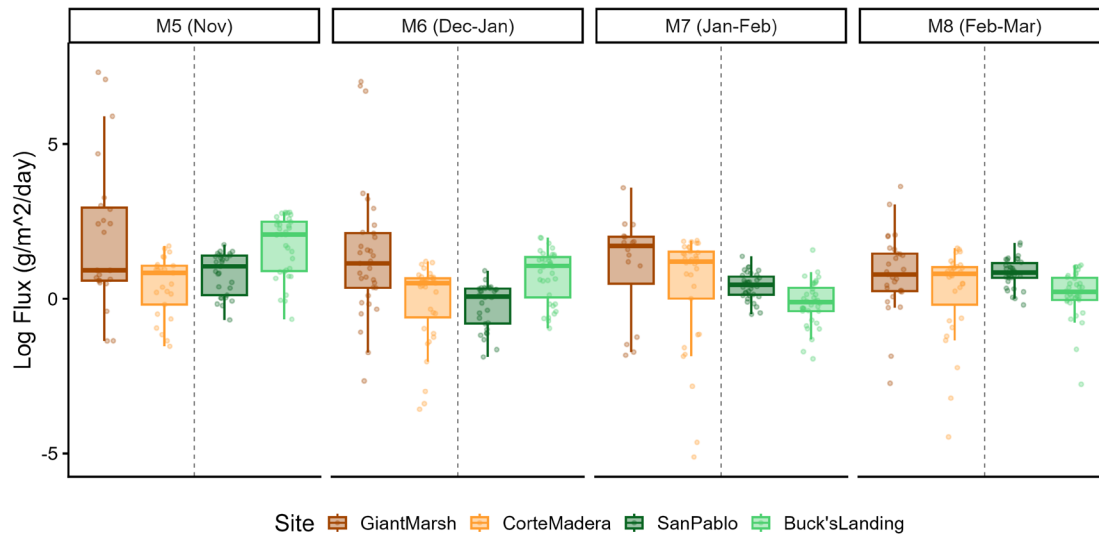


Figure 14. Sediment Deposition (Flux) Across March 2026 -June 2026 Within Ramped and Scarped Sites

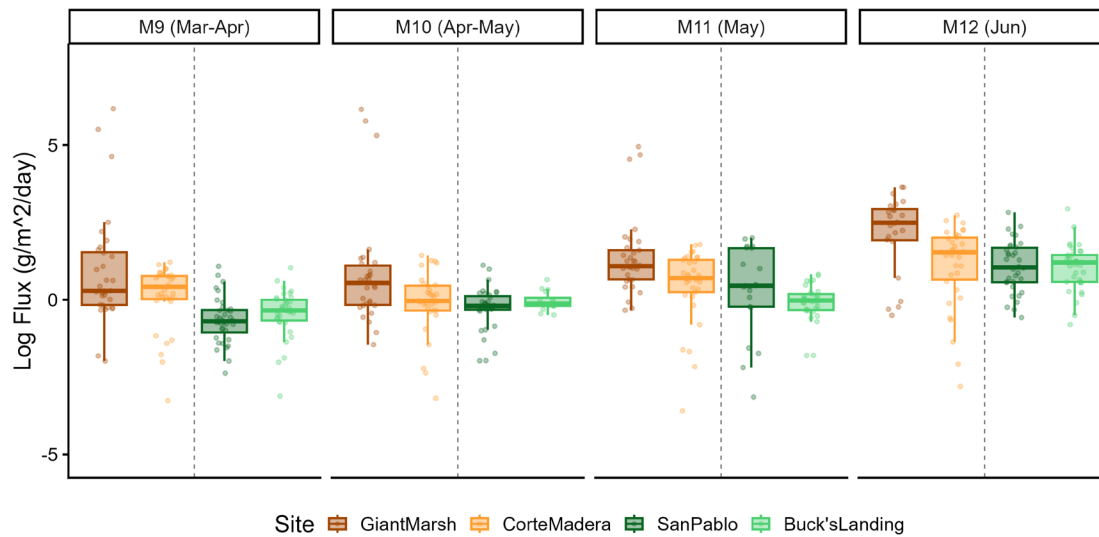


Table 9. Statistical Table of Sediment Deposition (Flux) Across Study Phase Collections (M#) within Corte Madera (2025-2026)

Study Phase	n	Mean	SD	SE	Median	Min	Max	Skewness	Kurtosis	Shapiro_p
M1	5	3.731	2.008	0.898	2.980	2.343	7.157	1.270	2.794	0.058
M2	10	2.904	1.816	0.574	1.843	0.855	6.082	0.600	1.847	0.084
M3	10	5.167	2.443	0.772	4.763	1.853	8.795	0.237	1.642	0.469
M4	10	1.959	2.118	0.670	1.200	0.073	6.614	1.265	3.318	0.026
M5	9	2.191	1.501	0.500	2.517	0.262	4.644	0.261	1.920	0.694
M6	12	1.429	0.802	0.232	1.598	0.037	2.630	-0.292	2.133	0.865
M7	12	2.951	2.067	0.597	3.751	0.006	5.888	-0.320	1.705	0.118
M8	12	2.244	1.594	0.460	2.260	0.026	5.128	0.328	2.362	0.509
M9	11	1.531	0.883	0.266	1.328	0.114	2.638	-0.229	1.827	0.303
M10	11	1.281	0.926	0.279	0.952	0.164	3.274	0.768	2.946	0.353
M11	11	2.378	1.509	0.455	2.094	0.110	5.137	0.307	2.427	0.883
M12	12	5.139	3.814	1.101	4.781	0.147	13.421	0.634	2.907	0.696

Flux units: g/m²/day

Kurtosis is RAW (not excess) -- a normal distribution reads ~3, not 0.

Shapiro_p < .05 indicates a significant departure from normality; NA where n < 3 or n > 5000.

Kruskal-Wallis across Study Phase within CorteMadera: statistic = 33.49, p < .001

Table 10. Statistical Table of Sediment Deposition (Flux) Across Study Phase Collections (M#) within Buck's Landing (2025-2026)

Study Phase	n	Mean	SD	SE	Median	Min	Max	Skewness	Kurtosis	Shapiro_p
M1	11	6.202	3.864	1.165	4.909	2.154	14.550	0.965	2.871	0.122
M2	11	2.861	1.460	0.440	2.649	1.203	5.320	0.572	1.942	0.180
M3	11	3.115	2.180	0.657	2.003	1.335	7.408	1.085	2.699	0.010
M4	11	2.229	1.306	0.394	1.480	0.688	4.116	0.229	1.323	0.054
M5	12	7.754	5.343	1.542	8.359	0.799	16.058	0.100	1.591	0.342
M6	12	2.818	1.759	0.508	2.959	0.495	6.125	0.297	2.177	0.753
M7	12	1.100	0.726	0.210	0.840	0.396	2.842	1.288	3.697	0.028
M8	12	1.419	0.754	0.218	1.234	0.278	2.740	0.577	2.454	0.322
M9	12	0.791	0.429	0.124	0.716	0.110	1.670	0.469	2.887	0.842
M10	6	1.048	0.348	0.142	0.990	0.687	1.674	0.987	2.829	0.348
M11	12	0.994	0.515	0.149	0.874	0.165	2.123	0.844	3.476	0.151
M12	12	3.702	2.090	0.603	3.721	0.786	7.455	0.412	2.287	0.641

Flux units: g/m²/day

Kurtosis is RAW (not excess) -- a normal distribution reads ~3, not 0.

Shapiro_p < .05 indicates a significant departure from normality; NA where n < 3 or n > 5000.

Kruskal-Wallis across Study Phase within Buck'sLanding: statistic = 70.82, p < .001

Table 11. Statistical Table of Sediment Deposition (Flux) Across Study Phase Collections (M#) within San Pablo (2025-2026)

Study Phase	n	Mean	SD	SE	Median	Min	Max	Skewness	Kurtosis	Shapiro_p
M1	8	11.039	11.438	4.044	6.379	2.824	32.448	1.159	2.564	0.006
M2	9	16.520	20.672	6.891	4.633	1.838	61.001	1.327	3.327	0.007
M3	10	14.569	12.235	3.869	8.094	2.776	34.683	0.503	1.554	0.033
M4	10	2.214	1.874	0.593	1.568	0.567	6.921	1.733	5.074	0.009
M5	10	2.671	1.427	0.451	3.194	0.886	4.703	-0.167	1.452	0.081
M6	10	0.984	0.545	0.172	1.058	0.268	1.917	0.143	1.861	0.424
M7	12	1.638	0.588	0.170	1.736	0.838	2.776	0.370	2.183	0.469
M8	12	2.483	1.183	0.341	2.415	1.044	5.278	1.057	3.601	0.163
M9	12	0.682	0.397	0.115	0.622	0.271	1.605	1.050	3.384	0.122
M10	12	0.919	0.405	0.117	0.836	0.282	1.671	0.501	2.834	0.245
M11	9	2.172	2.358	0.786	1.441	0.077	6.719	1.081	2.659	0.035
M12	12	3.883	2.515	0.726	3.169	0.852	8.902	0.822	2.555	0.191

Flux units: g/m²/day

Kurtosis is RAW (not excess) -- a normal distribution reads ~3, not 0.

Shapiro_p < .05 indicates a significant departure from normality; NA where n < 3 or n > 5000.

Kruskal-Wallis across Study Phase within SanPablo: statistic = 77.95, p < .001

Table 12. Statistical Table of Sediment Deposition (Flux) Across Study Phase Collections (M#) within Giant Marsh (2025-2026)

Study Phase	n	Mean	SD	SE	Median	Min	Max	Skewness	Kurtosis	Shapiro_p
M1	9	29.618	53.198	17.733	8.940	3.073	168.586	2.338	6.577	< .001
M2	10	109.764	217.870	68.897	8.493	1.262	596.904	1.628	3.759	< .001
M3	10	107.996	293.857	92.926	5.106	0.982	941.316	2.665	7.998	< .001
M4	9	126.903	363.186	121.062	2.987	0.800	1,095.227	2.512	7.120	< .001
M5	8	138.146	358.924	126.899	8.220	0.908	1,025.732	2.308	6.125	< .001
M6	11	93.138	291.384	87.856	4.371	0.779	971.590	2.874	9.096	< .001
M7	6	6.030	6.094	2.488	5.805	0.189	16.675	0.825	2.627	0.298
M8	10	4.487	6.259	1.979	2.599	0.111	21.312	2.222	6.511	< .001
M9	11	27.396	82.750	24.950	1.456	0.150	276.768	2.869	9.082	< .001
M10	10	34.823	104.598	33.077	1.822	0.467	332.509	2.700	8.110	< .001
M11	11	13.493	33.499	10.100	3.344	1.178	114.402	2.865	9.068	< .001
M12	9	14.032	10.194	3.398	12.334	0.711	35.596	0.818	3.419	0.332

Flux units: g/m²/day

Kurtosis is RAW (not excess) -- a normal distribution reads ~3, not 0.

Shapiro_p < .05 indicates a significant departure from normality; NA where n < 3 or n > 5000.

Kruskal-Wallis across Study Phase within GiantMarsh: statistic = 22.24, p = .023

Table 13. GLMM of Sediment Deposition (Flux) Across Summer (March-October) and Winter (November-February) Seasons Between Sites (2025-2026)

Term	Estimate	SE	z	p	exp(β)	95% CI low	95% CI high
(Intercept)	2.427	0.319	7.603	< .001	11.326	6.058	21.175
SiteSanPablo	-1.617	0.429	-3.766	< .001	0.199	0.086	0.461
SiteBuck'sLanding	-1.129	0.434	-2.603	.009	0.323	0.138	0.757
SiteCorteMadera	-1.818	0.428	-4.251	< .001	0.162	0.070	0.375
Cal_SeasonSummer	0.014	0.232	0.058	.953	1.014	0.643	1.598
SiteSanPablo:Cal_SeasonSummer	0.782	0.304	2.571	.010	2.187	1.204	3.971
SiteBuck'sLanding:Cal_SeasonSummer	-0.426	0.303	-1.406	.160	0.653	0.361	1.183
SiteCorteMadera:Cal_SeasonSummer	0.387	0.297	1.304	.192	1.472	0.823	2.632

Type II Wald χ^2 -- Site: $\chi^2(3) = 21.80$, $p < .001$

Type II Wald χ^2 -- Cal_Season: $\chi^2(1) = 4.76$, $p = .029$

Type II Wald χ^2 -- Site:Cal_Season: $\chi^2(3) = 20.92$, $p < .001$

Random intercept (Loc_ID): variance = 0.713, SD = 0.844 [48 groups]

N = 499 observations; AIC = 2493.8

Marginal $R^2 = 0.236$; conditional $R^2 = 0.638$ (NA = a random-effect variance was estimated at ~0)

Formula: Flux ~ Site * Cal_Season + (1 | Loc_ID)

Family: Gamma (log link) -- exp(β) = multiplicative change in Flux; 95% CI = Wald

4.2 Sediment Deposition Rates Across Years in One Ramped and One Scarped Edge Marsh

4.3 Sediment Deposition in Relation to Intra-Marsh Geomorphic Factors

5. Discussion

6. Conclusion